Solvency Field Theory: The Unified Framework (V9, Revised April 2026)

The Unified Framework

A Complete Information-Theoretic Foundation forDark Energy, Galaxy Dynamics, and General Relativity

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Position of this document within the framework

This document is the master framework presentation of Solvency Field Theory. Since its initial release, the framework has been extended through a focused series of companion notes that address specific structural questions left open in the V8 text. Readers new to SFT should read this master document first, then proceed to the companion notes in the order listed below. Several of the open questions identified in Section 14 of this document have since been addressed in the companion notes and are now referenced accordingly.

Companion note series (chronological):

- V8.1 — *The Recombination Handoff Companion Note.* Derives the H_0 tension contribution from the activation of independent Compton maintenance at recombination, treating decoupling as a fundamental bookkeeping transition rather than only a thermodynamic cooling event.
- V8.2 — *SFT, JWST Early Massive Galaxies, and Bottom-Up Structure Formation.* Resolves the JWST early-galaxy anomaly by applying SFT's weak-field enhancement to early diffuse baryonic clouds, producing bottom-up structure formation without top-down dark matter scaffolding.
- V8.4 revised — *Weak-Field SPARC Paper V2.* Upgrades the weak-field sector from an empirical fit to an internal, geometrically constrained result. Derives the rigid isotropic spherical-screen coefficient and the flat-disk axial-projection candidate, with morphology-dependent λ confirmed against SPARC at p = 0.0016.
- V8.5 — *Cosmological Sector Structural Advances.* Derives the forced four-state cosmic web ladder from axis factorization, the exact static projection corrections s_wall = 1/3 and s_filament = 2/3 from spherical-screen integrals, with TNG300 empirical confirmation at

two grid resolutions. Unifies Break #1 and Break #2 through a single projection-integral mechanism.

- **V8.6** — *Observer-Side Local-Environment Closure of the Cosmological Residual.* Closes the bounded w_a residual through a derived observer-side correction using path-averaged unscreened burden ΔU_rec(z), with covering-geometry isolation and a nine-point sensitivity sweep confirming robustness inside the pre-committed DESI closure radius.
- **V9** — *Weak-Field Lag Action and Repaired Scalar Gravity.* Records the weak-field lag-action closure through repaired Branch C*, which passes the classic scalar-gravity post-Newtonian gates (β = 1 exactly, first-order deflection $4GM/bc^2$), and documents the quantitatively constrained phase-gradient source-sector bridge as the present Goal 3 stopping point.

The pre-V8 documents (V6, V7.1 through V7.4, Phase III) remain in the repository as historical reference. Readers should treat them as background rather than as current reading, as much of their content has been superseded or integrated into the V8 series.

Abstract

We present an information-theoretic framework for cosmology built on a single physical postulate: every massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency $\nu = mc^2/h$, with each complete phase cycle demanding one Planck area of cosmological boundary. From this postulate, we derive a modified Friedmann equation differing from the standard result by a factor of exactly $1/(4\pi^2)$, a galactic acceleration threshold $a_{SFT} = cH_0/(2\pi)$ confirmed by 137 SPARC galaxies at 63% error reduction with zero free parameters, and a holographic count equation that serves as the native action principle of the theory. We demonstrate that the ratio $4\pi^2$ decomposes into two independent factors of 2π: one from the phase convention (linear cycles h versus angular radians ħ) and one from the boundary-versus-bulk normalization (Bekenstein-Hawking horizon accounting versus Einstein field equation coupling). This identity simultaneously explains the dark energy gap, the MOND acceleration scale, and the holographic capacity ratio. Applied to the cosmological expansion history using IllustrisTNG simulation data, the framework reproduces the evolving dark energy equation of state ($w_0 = -0.837$, $w_a = -0.615$) recently measured by DESI, preserves the Planck 2018 CMB acoustic peaks at 0.078% RMS, and recovers exact Schwarzschild geometry and the Tolman-Oppenheimer-Volkoff equations in the strong-field limit. The framework generates falsifiable predictions for environment-dependent structure growth (5–6% split between voids and walls at $z < 0.5$) testable by DESI, Euclid, and CMB-S4. Open problems include deriving the z-dependent screening evolution that produces cosmic acceleration and extending the constrained action beyond FLRW minisuperspace.

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1. Three Unsolved Problems

Three foundational puzzles in modern cosmology share a common feature: each involves an unexplained numerical scale that connects quantum mechanics to the large-scale structure of the universe. Despite decades of theoretical effort, none has a consensus explanation. This paper argues that all three are manifestations of a single geometric identity.

The dark energy problem. The observed cosmic acceleration requires an energy density parameterized as $\Omega_\Lambda \approx 0.69$ in the standard ΛCDM model. This number has no theoretical derivation. The naïve quantum field theory estimate of vacuum energy density exceeds the observed value by approximately 10^{120} orders of magnitude, making the cosmological constant problem one of the largest discrepancies in the history of physics. Standard cosmology treats Ω_Λ as a free parameter, manually adjusted to fit the data.

The MOND acceleration scale. Galaxy rotation curves deviate from Newtonian predictions below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10}$ m/s^2, as identified empirically by Milgrom [9] and confirmed across hundreds of galaxies over four decades. This scale is not predicted by any standard theory. Its approximate numerical coincidence with cH_0 has been noted repeatedly in the literature but never derived from a physical mechanism. The standard approach—invoking dark matter halos with tunable parameters—fits rotation curves but does not explain why a single universal acceleration scale governs the transition.

The holographic capacity gap. The Bekenstein-Hawking entropy of the cosmological horizon is approximately 10^{122} bits [1, 5]. The total entropy produced by all astrophysical processes over the entire history of the universe—including stellar evolution, black hole formation, and nucleosynthesis—amounts to approximately 10^{103} cumulative operations. The ratio between the boundary's information capacity and the bulk's processing history exceeds 10^{19}. No standard framework explains why the cosmological boundary appears so vastly over-provisioned relative to the physical processes it contains.

In this paper, we show that all three puzzles are manifestations of a single geometric identity: $(2\pi)^2 = 4\pi^2 \approx 39.48$. This factor arises from two independent unit conversions—one between linear and angular quantum phase, one between boundary thermodynamic normalization and bulk field equation coupling—and it appears whenever the Planck scale (ħ-defined) interfaces with the Compton scale (h-defined) within a cosmological geometry. We derive it from a single physical postulate about the information cost of quantum existence, confirm it against galactic rotation curve data, formulate a variational principle, and generate falsifiable predictions for forthcoming cosmological surveys.

2. The Compton Maintenance Postulate

A massive particle is not a static object. Its quantum state evolves continuously. The rate of this evolution is set by the particle's rest energy: higher mass means faster phase advance. Specifically, the quantum phase of any massive particle rotates at the Compton frequency:

$$\nu = mc^2 / h \quad (1)$$

where m is the particle's rest mass, c is the speed of light, and h is Planck's constant. This frequency is not a theoretical abstraction—it is the fundamental clock rate of the particle's quantum state, measurable through Compton scattering, matter-wave interferometry, and the de Broglie relation. A particle that stops evolving at its Compton frequency ceases to be a massive quantum object.

We propose a single physical postulate that connects this quantum fact to cosmological geometry:

The Compton Maintenance Postulate. *Every massive particle maintains its existence through continuous quantum phase evolution. Each complete phase cycle (2π radians, one linear Compton period) constitutes one irreversible bit of information, requiring one Planck area ($\ell_P^2 = \hbar G/c^3$) of cosmological boundary surface to accommodate it.*

Three distinct physical claims are embedded in this postulate, and each must be evaluated on its own merits:

Claim 1: Existence is computational. A massive particle persists by continuously processing information at its Compton frequency. This is not a metaphor. The phase evolution is a physical process with a definite rate, and each complete cycle represents an irreversible transition—the particle's state at the end of the cycle cannot be reversed to the state at the beginning without thermodynamic cost. This identification of quantum phase evolution with irreversible information processing is consistent with the Landauer principle [7], which establishes a minimum thermodynamic cost per bit of irreversible computation.

Claim 2: One bit equals one complete cycle. The fundamental unit of irreversible information is one complete phase revolution (2π radians), not one radian of phase advance. This is a specific, testable choice. If one bit corresponded to one radian instead of one full cycle, the theory would produce different numerical predictions—predictions that can be compared against galactic data. Section 4 presents this comparison and shows that the data prefer the complete-cycle convention.

Claim 3: The geometric cost is one Planck area per bit. Each irreversible bit demands one Planck area ($\ell_P^2 = \hbar G/c^3 \approx 2.61 \times 10^{-60}$ m^2) of cosmological boundary to accommodate it. This is consistent with the Bekenstein-Hawking entropy formula [1, 5], which assigns $S = A/(4\ell_P^2)$ bits of information to a surface of area A. The postulate promotes this from a statement about black hole thermodynamics to a process equation: information is continuously being written to the boundary, one Planck area at a time, by the phase maintenance of every massive particle in the universe.

3. Derivation of the Modified Friedmann Equation

3.1 Total information maintenance rate

If every massive particle processes one irreversible bit per Compton cycle, then the total information maintenance rate of the universe is the sum of all Compton frequencies. For a universe with total mass-energy M contained within the Hubble volume:

$$\dot{S} = Mc^2 / h \quad (2)$$

This is the total number of complete phase cycles performed by all massive particles per unit time. It has units of bits per second. Note that h (not ħ) appears because the postulate identifies one bit with one complete cycle.

3.2 Horizon information capacity

The information capacity of the cosmological boundary is given by the Bekenstein-Hawking formula applied to the Hubble horizon (radius $R_H = c/H$):

$$I = A_h / (4\ell_p^2) = 4\pi(c/H)^2 / (4\ell_p^2) = \pi c^2 / (\ell_p^2 H^2) \quad (3)$$

This is the total number of Planck-area cells on the Hubble horizon. It has units of bits. In flat FLRW spacetime, the Hubble sphere coincides with the apparent horizon—the surface that appears in rigorous cosmological thermodynamic derivations [6, 10]. The choice of this boundary is not arbitrary; it is the unique causal boundary within which information can be exchanged in one Hubble time.

3.3 The master equation

Solvency Field Theory identifies the Hubble parameter H as the fractional rate at which the cosmological boundary's information capacity is filled by bulk maintenance processing:

$$H = \dot{S} / I \quad (4)$$

This equation has a direct physical interpretation: the expansion rate of the universe is set by the ratio of how fast information is being produced (by ticking particles) to how much information the boundary can accommodate (Planck cells on the horizon). If the bulk processes more information, the boundary must be larger—the universe must expand more—to maintain the balance. The expansion rate H is the quantity that self-consistently satisfies this balance at every epoch.

3.4 Algebraic derivation

Substituting Eqs. (2) and (3) into Eq. (4):

$$H = [Mc^2/h] / [\pi c^2/(\ell_p^2 H^2)] \quad (5)$$

Express the total mass as $M = \rho V_H = \rho \times (4\pi/3)(c/H)^3$, and the Planck area as $\ell_P^2 = \hbar G/c^3 = Gh/(2\pi c^3)$:

$$H = [\rho \times (4\pi/3)(c^3/H^3) \times c^2/h] \times [Gh/(2\pi c^3) \times H^2/(\pi c^2)] \quad (6)$$

The factors of c^3, h, and one factor of π cancel. Collecting the remaining terms and multiplying both sides by H:

$$H^2 = 2G\rho / (3\pi) \quad (7)$$

This is the SFT modified Friedmann equation. It is derived from the Compton maintenance postulate with no free parameters, no fitting, and no ad hoc adjustments. It differs from the standard Friedmann equation $H^2 = 8\pi G\rho/3$ by a factor of:

$$\mathbf{H^2{}_s{}^{ft} / H^2{}_s{}^{td} = [2G\rho/(3\pi)] / [8\pi G\rho/3] = 1/(4\pi^2)} \quad (8)$$

The density ρ cancels in this ratio. The gravitational constant G cancels. What remains is a pure geometric factor: $4\pi^2 = (2\pi)^2 \approx 39.48$. This factor is content-independent—it does not depend on what is in the universe. Its origin and decomposition are the subject of Section 5.

4. The 2π Factor and Galactic Confirmation

4.1 The acceleration threshold

Taking the square root of the ratio in Eq. (8):

$$\mathbf{H_s{}^{ft} = H_s{}^{td} / (2\pi)} \quad (9)$$

The natural acceleration scale of the Compton maintenance mechanism is therefore:

$$\mathbf{a_s{}^{ft} = cH_0 / (2\pi) \approx 1.05 \times 10^{-10}\ m/s^2} \quad (10)$$

This number is derived, not fitted. The factor of 2π is the ratio between the linear Compton frequency $\nu = mc^2/h$ (complete cycles per second) and the angular Compton frequency $\omega = mc^2/\hbar$ (radians per second). It enters the theory because the postulate identifies one irreversible bit with one complete phase cycle, not one radian.

The predicted value $a_{SFT} = 1.05 \times 10^{-10}$ m/s^2 is 13% below MOND's empirical acceleration threshold $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 [9], which was identified from galactic data without any theoretical derivation. The agreement to 13% from zero free parameters is striking, but the data must decide whether $cH_0/(2\pi)$ specifically—as opposed to some nearby scale—correctly describes the weak-field transition.

4.2 SPARC rotation curve test

We test $a_{SFT} = cH_0/(2\pi)$ against 137 quality-filtered galaxies from the SPARC database [8], which provides observed rotation curves and baryonic mass profiles derived from 3.6 μm Spitzer photometry for 175 nearby disk galaxies. After applying quality filters (inclination > 30°, distance < 100 Mpc, sufficient radial coverage), 137 galaxies remain.

For each galaxy, we compute the predicted rotation curve using only the observed baryonic mass profile and a single universal acceleration scale. The predicted gravitational acceleration at each radius is:

$$\mathbf{g_{pre}{}^d = g^b{}_{ar} \times \nu(g^b{}_{ar} / a_s{}^{ft})} \quad (11)$$

where g_{bar} is the Newtonian gravitational acceleration from baryonic mass, $a_{SFT} = cH_0/(2\pi)$, and $\nu(x) = (1 - e^{-\sqrt{x}})^{-1}$ is an exponential interpolation function. No per-galaxy parameters are adjusted. The only input is the observed baryonic mass profile; the only theoretical parameter is the universal scale a_{SFT}.

To evaluate the significance of the 2π factor, we compare four acceleration scales:

Model	Mean relRMSE	Improvement
Baryons only (no correction)	0.458	—
SFT: $a = cH_0/(2\pi) = 1.05 \times 10^{-10}$	**0.167**	**63% reduction**
Alternative: $a = cH_0$ (no 2π)	0.283	38% reduction
Alternative: $a = cH_0/6 = 1.13 \times 10^{-10}$	0.174	62% reduction

Table 1: Rotation curve error across 137 SPARC galaxies. The theoretically derived scale $cH_0/(2\pi)$ outperforms all alternatives, including the literal scale cH_0. The interpolation function and quality filters are identical across all four models; only the acceleration scale differs.

The SPARC data confirm two independent claims. First, that the Hubble-scale constant cH_0 sets the relevant weak-field acceleration scale—eliminating baryons-only models by a factor of 2.7 in error. Second, that the specific factor of 2π is physically real—the derived scale $cH_0/(2\pi)$ outperforms the literal cH_0 by a factor of 1.7 in error. The data independently select the 2π factor that the Compton maintenance postulate predicts.

The close performance of $cH_0/6$ (relRMSE 0.174 versus 0.167 for $cH_0/(2\pi)$) reflects the proximity of $2\pi \approx 6.28$ to 6.0. The data prefer the derived value but do not exclude nearby alternatives at high statistical significance with this sample size. Future high-precision surveys (e.g., WALLABY, SKA) with larger galaxy samples and lower systematic uncertainties will provide a sharper test.

The 13% discrepancy between $a_{SFT} = 1.05 \times 10^{-10}$ m/s^2 and MOND's empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 has several possible origins: QCD binding energy contributing to the effective Compton frequency (the proton's mass is approximately 99% binding energy), cosmological evolution of H_0 between the epoch when a_0 was empirically calibrated and z = 0, systematic uncertainties in the empirical a_0 determination, or the choice of interpolating function. The gap is noted as an open problem (Section 14).

5. The $4\pi^2$ Unification

The factor $4\pi^2 = (2\pi)^2 \approx 39.48$ is not a single unexplained constant. It appears in four distinct physical contexts, and in each case it decomposes into the same two independent factors of 2π: one from the phase convention used to define one bit of information, and one from the different normalization conventions used by holographic boundary thermodynamics and bulk gravitational field equations to relate geometry to matter content. The first factor enters through the Compton maintenance postulate; the second enters through the structure of the Bekenstein-Hawking formula and the Einstein field equations. Neither depends on the matter content of the universe.

5.1 The Friedmann ratio

As derived in Eq. (8), $H^2_{SFT} / H^2_{Friedmann} = 1/(4\pi^2)$. The SFT expansion rate is $4\pi^2$ times smaller than the standard Friedmann rate for the same matter density. The first 2π arises because the master equation uses $\nu = mc^2/h$ (complete cycles) where Friedmann's derivation implicitly uses $\omega = mc^2/\hbar$ (angular frequency). The second 2π arises because the boundary capacity $I = A/(4\ell_P^2)$ uses the Bekenstein-Hawking normalization while the Friedmann equation uses the Einstein coupling $8\pi G$, which includes a different geometric prefactor.

5.2 The galactic acceleration scale

Taking the square root of the $4\pi^2$ ratio: $a_{SFT} = cH_0/(2\pi)$, not cH_0. The single factor of 2π appears at the galactic scale because the acceleration threshold scales as H (linearly), not as H^2. Acceleration involves the square root of the cosmological ratio, so only one factor of 2π is exposed. This is why MOND's empirical a_0 is close to $cH_0/(2\pi)$ rather than cH_0 or $cH_0/(4\pi^2)$.

5.3 The holographic capacity ratio

The total number of Compton phase cycles performed by all mass-energy in the observable universe over its entire history is:

$$\mathbf{N_{ti}c^k{}_s = Mc^2 / (hH_0) \approx 5.7 \times 10^{120}} \quad (12)$$

The total number of Planck-area cells on the Hubble horizon is:

$$\mathbf{N^b{}_{its} = A / (4\ell_p{}^2) \approx 2.2 \times 10^{122}} \quad (13)$$

Their ratio is:

$$\mathbf{N_{ti}c^k{}_s / N^b{}_{its} = 1/(4\pi^2) = 0.0253} \quad (14)$$

The holographic boundary has exactly $4\pi^2$ times more capacity (measured in ħ-based Planck cells) than the total processing history of the universe (measured in h-based Compton cycles). If the ledger were restated in consistent units—either all angular or all linear—the apparent excess disappears. The "holographic over-provisioning" is a bookkeeping artifact of using angular units for a linear process.

5.4 The dark energy gap

The Compton maintenance equation with matter-only density gives $H_{SFT} = 0.56 \times H_{obs}$ (56% of the observed expansion rate, since $\sqrt{(2/3\pi \times \Omega_m)} \approx 0.56$ when $\Omega_m = 0.31$). The "dark energy gap"—the remaining 44% of the observed expansion rate—is structured by the same $4\pi^2$ geometric identity that appears in the three previous contexts. Whether the $4\pi^2$ identity fully accounts for the gap or whether additional physics (z-dependent screening, structural channels) is required to produce the observed acceleration is addressed in Sections 10–11.

6. The Rigorous Decomposition of $4\pi^2$

In V7.1, the $4\pi^2$ was attributed to "one factor of 2π from the frequency conversion, and a second from projecting flat-space results onto a spherical boundary." That attribution was directionally correct for the first factor but imprecise for the second. An earlier conjecture attributed the full $4\pi^2$ to a ratio between locally-sourced and globally-sourced expansion contributions; this was shown to be inconsistent with Birkhoff's theorem and has been rejected. This section provides the corrected, rigorous decomposition.

6.1 Factor 1: Phase convention (h vs ħ)

The count equation (Section 8) uses h in the bulk maintenance rate and ħ in the Planck area. Since $h = 2\pi\hbar$, this introduces one factor of 2π. To isolate it, consider replacing h with ħ in the bulk count—counting one tick per radian of phase advance rather than one tick per complete cycle. The resulting Friedmann-type equation becomes:

$$\mathbf{H^2 = 4G\rho / 3} \quad (15)$$

This still differs from standard Friedmann ($H^2 = 8\pi G\rho/3$) by a factor of 2π, confirming that only one of the two factors originates from the phase convention. The first 2π is: **complete cycles (h) versus angular phase (ħ)**. It is empirically confirmed by the SPARC test (Section 4).

6.2 Factor 2: Boundary vs bulk normalization

The remaining factor of 2π is independent of the phase convention. It arises from the different normalization conventions used by boundary thermodynamics and bulk field equations.

On the SFT/holographic side, the boundary capacity is $N_H = A/(4\ell_P^2)$, where the factor of 4 comes from the Bekenstein-Hawking entropy formula [1, 5]. This is a thermodynamic normalization that sets the number of independent degrees of freedom on a gravitational horizon.

On the GR/bulk side, the Einstein field equation carries a coupling coefficient $8\pi G$:

$$\mathbf{G\mu_\nu = 8\pi G\, T\mu_\nu} \quad (16)$$

The 8π is not derived from first principles within GR—it is fixed by requiring consistency with the Newtonian limit via the Poisson equation $\nabla^2\Phi = 4\pi G\rho$. The ratio between these two normalization conventions—how SFT counts boundary cells versus how GR couples bulk geometry to matter—introduces the second factor of 2π.

6.3 The complete factorization

$$\mathbf{4\pi^2 = 2\pi\ (\text{phase convention}) \times 2\pi\ (\text{boundary/bulk normalization})} \quad (17)$$

Factor	Origin	SFT uses	GR uses

First 2π	Phase convention: what counts as one irreversible bit	h (complete cycles)	ħ (radians)
Second 2π	Geometry coupling: how content maps to geometry	$A/(4\ell_p^2)$ boundary	8πG bulk coupling
Product	Total: H^2_Friedmann / H^2_SFT	$\mathbf{4\pi^2 \approx 39.48}$	*(content-independent)*

Table 2: The two-factor decomposition of $4\pi^2$. Each factor has a distinct and independently identifiable origin. Neither depends on the matter content of the universe. The density ρ and gravitational constant G cancel completely in the ratio.

7. The Holographic Count Equation

7.1 Motivation

The algebraic derivation in Section 3 is correct, but it does not constitute a variational principle. It does not specify the mathematical object that the theory extremizes, nor does it explain why the Hubble time enters as the natural settlement interval. This section derives the same result from a more fundamental starting point: a count equality between the boundary and the bulk.

7.2 The postulate

The Holographic Count Equation. *The number of Planck-area cells on the Hubble horizon equals the number of complete Compton phase cycles performed by all bulk matter during one Hubble maintenance interval:*

$$\mathbf{A_h / (4\ell_p^2) = E_h / (hH)} \quad (18)$$

The left side counts boundary cells. The right side counts bulk processing cycles accumulated over one Hubble time $t_H = 1/H$. The equation states that these are equal: the information capacity of the boundary exactly matches the information processed by the bulk over one causal maintenance interval.

The Hubble time is not an arbitrary choice. It is the timescale over which a signal can traverse the Hubble radius c/H. Beyond this interval, the ledger cannot be causally settled. The count equation is therefore an apparent-horizon thermodynamic balance: the boundary refreshes its information budget every Hubble time, and the bulk fills it exactly.

7.3 Derivation

In flat FLRW with homogeneous density ρ:

Left-hand side. Using $\ell_P^2 = \hbar G/c^3$:

$$\mathbf{N_h = \pi c^5 / (\hbar G H^2)} \quad (19)$$

Right-hand side. Using $E_H = (4\pi/3)\rho c^5/H^3$ and $h = 2\pi\hbar$:

$$\mathbf{N_{ti}c^k = 2\rho c^5 / (3\hbar H^4)} \quad (20)$$

Equating. Setting $N_H = N_{tick}$, cancelling c^5 and $\hbar$:

$$\mathbf{\pi/(GH^2) = 2\rho/(3H^4)} \quad (21)$$

Multiplying both sides by GH^4 and dividing by π:

$$\mathbf{H^2 = 2G\rho / (3\pi)} \quad (22)$$

This is the same modified Friedmann equation derived algebraically in Eq. (7), now obtained from the holographic count equation with zero free parameters and no proportionality constant.

7.4 Why the instantaneous rate law required $\gamma = 3$

An earlier attempt formulated the SFT law as an instantaneous rate equation: $d/dt(A_H/4\ell_P^2) = \gamma \times E_H/h$. This produced $H^2 = 2\gamma G\rho/(9\pi) + C$, requiring $\gamma = 3$ inserted by hand to recover the V7.1 coefficient.

The count equation resolves this by recognizing that the theory has **memory over one causal interval**. The information ledger is not settled instantaneously; it is settled over one Hubble time. The factor of $1/H$ in the count equation (through the maintenance interval $t_H = 1/H$) carries exactly the volume-to-area geometric information that the rate law was missing. The factor of 3 was never an unexplained constant—it was always implicit in the relationship between the Hubble sphere's volume (where ticking happens) and its surface area (where capacity is measured). The count formulation lets this factor emerge from the algebra instead of requiring it as an input.

8. The Constrained Action Principle

The holographic count equation can be promoted to a variational principle by introducing a Lagrange multiplier $\Lambda(t)$ that enforces the ledger constraint:

$$\mathbf{S_eff = \int dt\, \Lambda(t)\, [A_h(a,H)/(4\ell_p^2) - E_h(a,\rho)/(hH)] + S_{matter}} \quad (23)$$

where $a(t)$ is the FLRW scale factor, $H = \dot{a}/a$, and S_{matter} is the standard matter action ensuring the continuity equation $\dot{\rho} + 3H\rho = 0$ for pressureless dust.

This is a **constrained minisuperspace action**. It is not a standard local Lagrangian density integrated over spacetime. It is a holographic constraint enforced at the boundary level, relating the rate of boundary growth to the rate of bulk information processing. The fundamental law of the theory is a balance between surface capacity and volume content—consistent with the holographic and information-theoretic character of the framework.

The Lagrange multiplier $\Lambda(t)$ plays the role of a maintenance cost enforcer—it measures the cost of violating the count balance. At the classical level, Λ is determined by the constraint itself and does

not introduce independent dynamics. This is directly analogous to the role of the lapse function in canonical general relativity, where the Hamiltonian constraint is enforced by a multiplier without generating new physical degrees of freedom.

Formulation	Equation	Free parameters	Produces Eq. (7)?
Master equation (V7.1)	$H = \dot{S}/I$	None	Yes (algebraic)
Instantaneous rate law	$d/dt(N_h) = \gamma \times \dot{N}_{ti}c^k$	$\gamma = 3$ (hand-set)	Only if $\gamma = 3$
Holographic count equation	$\mathbf{N_h = N_{ti}c^k}$ **(Eq. 18)**	**None**	**Yes (derived)**
Constrained action	$S_e$ff with $\Lambda(t)$ (Eq. 23)	None	Yes (variational)

Table 3: Four formulations of the SFT cosmological law, listed in order of increasing mathematical maturity. The holographic count equation and the constrained action principle are the native formulations; both produce the modified Friedmann equation with zero free parameters.

9. The Planck Scale as Minimax Solvent Resolution

The appearance of the Planck area ℓ_P^2 in the count equation is not simply assumed from the Bekenstein-Hawking formula. It can be independently motivated as the unique scale at which quantum localization and gravitational confinement are simultaneously satisfiable.

For a localized excitation of mass m, two fundamental bounds apply. The quantum localization bound requires $L_Q \geq \hbar/(mc)$—a configuration cannot be localized below its reduced Compton wavelength without requiring energies that produce pair creation. The gravitational confinement bound requires $L_G \geq Gm/c^2$—a configuration cannot be compressed below its Schwarzschild radius without forming a black hole.

A physical configuration is solvent—simultaneously quantum-mechanically localizable and gravitationally stable—only if its size L satisfies $L \geq \max(L_Q, L_G)$. Minimizing the required size over all possible masses m yields $L_Q = L_G$ at:

$$\mathbf{m^* = m_p = \sqrt{(\hbar c/G)}, \quad L^* = \ell_p = \sqrt{(\hbar G/c^3)}} \qquad (24)$$

The Planck scale is the **minimax solvent resolution**: the smallest cell size at which both quantum and gravitational consistency constraints are simultaneously satisfied. Below the Planck scale, the two bounds conflict—no solvent configuration exists. At the Planck scale, they meet. Any information-theoretic framework that claims to tile the boundary with the minimal resolution consistent with both quantum mechanics and gravity must tile it in Planck-area cells.

This derivation provides a first-principles justification for the boundary cell size in Eq. (18), independent of the Bekenstein-Hawking formula. The Planck area appears in SFT not because it is borrowed from black hole thermodynamics, but because it is the unique scale where the universe's quantum and gravitational accounting systems agree.

10. The Gravitational Screening Mechanism

The Compton maintenance postulate implies that the geometric cost of each quantum phase cycle must be accommodated somewhere in the spacetime geometry. This cost can be settled in one of two ways, depending on the local gravitational environment.

Inside gravitational wells (galaxies, galaxy clusters, filaments): the pre-existing spacetime curvature—gravity itself—absorbs the geometric maintenance cost. The particle's information budget is settled locally against the curved geometry that already exists. This is the screened regime. The expansion contribution per particle is minimal because the local gravitational structure has already provided the geometric infrastructure needed to accommodate the tick. This is why the bare acceleration threshold $a_{SFT} = cH_0/(2\pi)$ correctly governs galactic dynamics: inside galaxies, the cost is settled at the local 1:1 rate, one Planck area per complete phase cycle.

In cosmic voids (which constitute 99.96% of the total cosmic volume by the IllustrisTNG TNG300 halo catalog): there is no local spacetime curvature to absorb the maintenance cost. Each particle's full geometric price—including the $4\pi^2$ conversion between angular boundary cells and linear phase cycles—is transmitted to the cosmological boundary. Expansion in voids is maximally efficient per unit tick.

This environmental dependence is not a free parameter. It follows directly from the postulate: gravity absorbs geometric cost locally; absence of gravity transmits it globally. The quantitative threshold between the screened and unscreened regimes is set by $a_{SFT} = cH_0/(2\pi)$—the acceleration below which local curvature can no longer fully absorb the maintenance cost, and cosmic-scale effects become dynamically relevant.

The screening mechanism provides a unified physical explanation for three phenomena that are conventionally treated as unrelated: (1) the MOND transition in galaxies (where local screening gives way to cosmological maintenance), (2) the dark energy gap in cosmology (where unscreened voids dominate the expansion budget), and (3) the connection between the MOND scale and the Hubble constant (both are set by the same maintenance physics). In the SFT framework, these are not three phenomena—they are three manifestations of one mechanism viewed at different scales.

11. The Bimodal Entropy Ledger

11.1 Data source and channel structure

To connect the Compton maintenance postulate to the observed expansion history, we model the entropy production history of the universe using the IllustrisTNG TNG300 cosmological simulation [11]. This simulation provides a cubic volume of $(205\ \mathrm{Mpc/h})^3$ containing approximately 700,000 black holes tracked across 100 snapshots from z = 20 to z = 0, along with detailed halo catalogs, star formation histories, and merger trees.

A methodological note is warranted. IllustrisTNG TNG300 is a ΛCDM simulation that reproduces observed gravitational structure — rotation curves, gravitational lensing, cluster dynamics, and the

cosmic web — using cold dark matter particles as a computational substrate. SFT interprets the gravitational organization captured by these simulations differently: what ΛCDM attributes to dark matter halos, SFT attributes to gravitational screening infrastructure that emerges from the information-maintenance balance. The numerical content, however, is the same. TNG300 is calibrated against observation, and the gravitational potentials it produces are empirically accurate regardless of the ontological interpretation assigned to the mass. When SFT extracts M200c from the simulation, it is measuring organized gravitational architecture — the specific quantity that determines local screening strength — not endorsing the existence of dark matter particles. The structural specificity test (Section 11.2) confirms this interpretation: virialized gravitational organization (M200c) succeeds where raw mass inventory fails, indicating that the relevant physical quantity is the degree of gravitational organization, not the total amount of mass or its particle identity. No native SFT simulation currently exists. When one is built, the gravitational structure it produces would differ in its microphysical origin but not in its observable gravitational effects — which are the quantities the bimodal model actually uses.

The astrophysical processing within this volume decomposes naturally into two channels:

The active processing channel measures the rate of ongoing astrophysical transactions: black hole accretion rates and star formation rates. These represent irreversible information-producing events—the Compton ticks associated with matter actively converting gravitational potential into thermal radiation, nuclear binding energy, or relativistic jets. The active channel captures the *transactional* component of the universe's information budget.

The passive structural capacity channel measures the accumulated gravitational architecture: the virialized halo mass M_{200c} summed over all halos in each snapshot. This represents the organized gravitational structure that has already been built and is now maintaining itself—the screening infrastructure that absorbs local maintenance costs. The passive channel captures the *occupancy* component: the geometric cost of maintaining existing structure, not the cost of building new structure.

The necessity of two channels is not assumed—it is empirically demonstrated. An active-only model (using BH accretion and star formation alone) systematically underpredicts the dark energy density at low redshift. A passive-only model lacks the dynamical evolution needed to produce the observed phantom crossing. The bimodal combination succeeds where neither channel alone does.

(The bimodal active/passive decomposition presented here has been generalized in the V8.5 companion note to a four-state cosmic web ladder — void, wall, filament, halo — with forced state weights from axis factorization and static projection corrections derived from spherical-screen integrals. The two descriptions are consistent in the appropriate limits, but the V8.5 framework supersedes the original bimodal presentation as the more complete structural picture. The bimodal text is retained here as the original V8 derivation path.)

11.2 The structural specificity test

A critical diagnostic distinguishes the passive channel from a generic dark matter proxy. When the passive channel is defined using raw dark matter halo mass M_{halo} instead of the virialized mass M_{200c},

the resulting equation of state shifts to $w_0 = -1.065$—overshooting the phantom divide and missing the observed value. The virialized mass M_{200c}—which specifically measures gravitationally organized, thermally relaxed structure—succeeds where raw inventory fails.

This structural specificity is physically meaningful within the SFT framework. The maintenance cost that matters is not the total amount of mass but the amount of mass that has been gravitationally organized into screening infrastructure. Unvirialized dark matter contributes to the gravitational inventory but does not participate in the same way in the information-maintenance balance. The distinction between M_{200c} and M_{halo} is a physical discriminator that no other dark energy model predicts.

11.3 The dark energy equation of state

The bimodal entropy production history, mapped onto a CPL parameterization, yields:

$$\mathbf{w_0 = -0.837, \quad w_a = -0.615} \qquad (25)$$

This lands at a distance of 0.005 in the CPL (w_0, w_a) plane from the Pantheon+ supernova best-fit. The DESI DR2 baryon acoustic oscillation measurement [2] independently favors evolving dark energy with $w_0 > -1$ and $w_a < 0$, consistent with the SFT bimodal prediction.

The bimodal model involves no additional cosmological fitting beyond the channel identification itself. The active and passive channels are extracted directly from the simulation data. Their relative weighting and redshift evolution are determined by the astrophysics of the simulation, not by tuning to match dark energy observations. However, the channel structure is empirically identified from TNG300 data, not derived from the Compton maintenance postulate. The identification of the channels with specific physical processes (transactional processing and structural occupancy) is motivated by the postulate but not yet formally derived. This connection is the subject of Section 11.5.

11.4 CMB compatibility

The bimodal expansion history, parameterized as a CPL dark energy model, was tested through the CLASS Boltzmann solver [11] against the Planck 2018 CMB temperature power spectrum. The resulting RMS residual across all multipoles (ℓ = 2–2500) is:

$$\mathbf{RMS\ residual = 0.078\%} \qquad (26)$$

By multipole regime: the ISW sector (ℓ = 2–30) shows a 1.8% residual, consistent with cosmic variance at low multipoles. The first acoustic peak (ℓ = 30–300) shows 0.104%. The acoustic damping tail (ℓ = 300–2500) shows 0.163%. All are within or near the precision of the Planck measurement itself.

This compatibility is expected on physical grounds. At the CMB epoch ($z \approx 1100$), the universe contains no virialized halos, no galaxy clusters, no voids, and no gravitational screening infrastructure. The Compton maintenance term is subdominant because the universe is radiation-dominated and no organized structure exists to create differential screening. SFT's

modifications emerge only at late times as massive structures form and the screening mechanism activates. The framework preserves standard early-universe physics by construction.

11.5 Connecting the count equation to the bimodal model

This section addresses the central structural question of the unified framework: how does the holographic count equation (Section 7), which produces $H^2 = 2G\rho/(3\pi)$ with constant parameters, connect to the bimodal model (Sections 11.1–11.4), which produces a z-dependent w(z) with evolving channels?

The problem. The count equation with constant parameters gives an effective equation of state w = 0—the same as matter. It produces the correct H_0 amplitude but does not produce cosmic acceleration ($w < -1/3$). This was verified by testing a constant-parameter version through CLASS, which confirmed that the resulting w(z) is flat at zero across all redshifts. The count equation alone, with time-independent parameters, cannot explain why the universe transitioned from deceleration to acceleration.

The resolution. Cosmic acceleration requires the count balance to evolve with redshift. The physical mechanism for this evolution is the formation of structure over cosmic time. Between z = 2 and z = 0, the universe underwent a dramatic transformation: the first massive halos collapsed, galaxy clusters formed, the cosmic web differentiated into filaments, walls, and voids, and the fraction of cosmic volume occupied by screening infrastructure changed fundamentally.

In the SFT framework, this structural evolution modifies the count balance in two ways:

The active processing rate declines. Both black hole accretion and star formation peak near $z \approx 2$ and decline steeply toward z = 0. The universe's transactional information production is slowing down. In count equation terms, the bulk maintenance rate $\dot{S} = Mc^2/h$ is decreasing—not because mass is disappearing, but because the fraction of mass actively converting potential energy into radiation is declining as fuel is exhausted.

The structural screening fraction grows. The total virialized halo mass M_{200c} increases monotonically from z = 2 to z = 0. More structure means more local screening, which means a larger fraction of the total Compton ticking is absorbed locally rather than transmitted to the cosmological boundary. The effective cosmological maintenance cost per unit mass decreases as more mass becomes gravitationally organized.

The combination of declining active processing and growing passive screening produces a net shift in the effective equation of state from $w \approx 0$ (matter-dominated, minimal screening, high active processing) toward $w < -1/3$ (screening-dominated, low active processing, structural occupancy driving the residual expansion). The bimodal model captures this evolution empirically through the TNG300 simulation data.

Current status. The count equation provides the mathematical foundation: boundary cells equal bulk ticks over one Hubble time. The bimodal model provides the z-dependent architecture. The quantitative evolution of the screening fraction with redshift has since been derived in the V8.5 cosmological sector companion note, which established the forced four-state ladder ($w_void = 4\pi^2$,

w_wall = (2π)^(4/3), w_filament = (2π)^(2/3), w_halo = 1) from axis factorization, and derived the exact static projection corrections s_wall = 1/3 and s_filament = 2/3 from spherical-screen second-moment integrals, with direct TNG300 empirical confirmation at two independent grid resolutions. The remaining low-redshift residual was subsequently closed in the V8.6 observer-side local-environment companion note through a derived covering-geometry correction, with robustness confirmed across a nine-point sensitivity sweep inside the pre-committed DESI closure radius. The bimodal decomposition presented in the subsequent sections is consistent with the V8.5 ladder framework as its coarse-grained limit, but readers interested in the full derivation should consult V8.5 and V8.6 directly.

12. Strong-Field Closure: Exact General Relativity

12.1 The local compactness parameter

At the scale of compact objects—neutron stars, black holes, and dense stellar remnants—the Compton maintenance postulate can be applied locally rather than cosmologically. The relevant parameter is the compactness:

$$\chi = GM / (2c^2 r) \quad (27)$$

which measures the ratio of gravitational potential energy to rest-mass energy at radius r from a mass M. For $\chi \ll 1$, the system is in the weak-field regime (galaxies, diffuse structure). For $\chi \to 1/2$, the system approaches the Schwarzschild limit.

12.2 Recovery of the Schwarzschild metric

When the local analog of the master equation is applied to a spherically symmetric, static mass distribution, the resulting metric is:

$$ds^2 = -(1 - 2GM/c^2 r)c^2 dt^2 + (1 - 2GM/c^2 r)^{-1} dr^2 + r^2 d\Omega^2 \quad (28)$$

This is the exact Schwarzschild exterior metric. It is not an approximation, not a leading-order expansion, and not a modification of general relativity. The information-maintenance balance, applied locally, produces the same geometry that Einstein's field equations produce when solved for a point mass in vacuum.

12.3 Recovery of the TOV equations

For the interior of a self-gravitating fluid sphere, the same local maintenance balance produces the Tolman-Oppenheimer-Volkoff hydrostatic equilibrium equations, which govern the structure of neutron stars and other compact objects. The pressure gradient required to support a fluid element against gravitational collapse is exactly the gradient that the maintenance balance requires to keep the information budget solvent at every radius.

12.4 Interpretation

SFT does not modify general relativity. It provides an information-theoretic *foundation* for it. In the strong-field regime ($\chi \to 1$), the local maintenance cost dominates all other contributions. The system has no choice but to adopt the geometry that exactly balances its information budget—and that geometry is Schwarzschild. General relativity emerges as the strong-field limit of the information-maintenance principle, just as Newtonian gravity emerges as the weak-field limit.

The strong-to-weak transition is the screening mechanism itself. At high compactness, maintenance is fully absorbed locally—the system is completely screened, and standard GR governs. At low compactness, local screening is insufficient, and the cosmological maintenance cost becomes dynamically relevant—producing the MOND-like weak-field behavior measured in Section 4. The screening mechanism is not an add-on; it is the bridge between the two regimes of the same underlying principle.

13. Falsifiable Predictions

The gravitational screening mechanism produces four quantitative predictions that are unique to environment-dependent dark energy models and cannot be replicated by ΛCDM with a spatially uniform cosmological constant. These predictions follow from the screening mechanism with no additional free parameters.

Prediction	ΛCDM	SFT	Observatory
Growth rate split: $f\sigma_8$ in voids vs walls at $z < 0.5$	Zero	**5–6%**	DESI, Euclid
ISW signal in supervoids	Standard	**Enhanced**	CMB-S4, Planck
Mean $f\sigma_8$ vs ΛCDM at $z \sim 0.3$	Baseline	1–2% lower	DESI DR3
Void expansion rate	Same as mean	**Faster than mean**	BOSS, DESI

Table 4: Falsifiable predictions distinguishing SFT from ΛCDM. All are zero-parameter consequences of the screening mechanism. ΛCDM predicts exactly zero environment dependence because Λ is spatially uniform.

The growth rate predictions are computed by solving the standard linear growth equation $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho\delta$ with the effective expansion rate modified by the local screening strength. In overdense environments, more gravitational curvature absorbs more of the maintenance cost, reducing the effective expansion rate and enhancing structure growth. In underdense environments (voids), less screening produces faster effective expansion and suppressed growth. The predicted 5–6% split at $z < 0.5$ between overdensities ($\delta \approx +0.5$) and deep voids ($\delta \approx -0.5$) is within the projected sensitivity of DESI DR3 and Euclid.

The ISW prediction is directly relevant to existing data. Granett, Neyrinck, and Szapudi [4] measured anomalously strong Integrated Sachs-Wolfe signals from supervoids, exceeding ΛCDM expectations by a factor of 2–3. This anomaly has persisted through multiple independent analyses and has no satisfactory explanation within ΛCDM. SFT's screening mechanism provides a natural explanation: supervoids have less gravitational screening, so their gravitational potentials decay

faster than a uniform Λ would predict, producing a stronger ISW imprint on CMB photons traversing those regions.

14. Open Opportunities

The framework presented here makes specific claims, produces confirmed predictions, and generates testable forecasts. It also identifies directions where the framework has room to extend. This section catalogs them honestly. Several items originally listed here as open problems have since been addressed by the V8.x companion note series and V9, and are noted accordingly.

14.1 Cosmic acceleration from the count equation. *Closed in V8.5 and V8.6.* The z-dependent screening evolution was derived in V8.5 through the forced four-state ladder and exact static projection corrections, with TNG300 empirical confirmation. The remaining low-redshift w_a residual was closed in V8.6 through the observer-side local-environment channel, with closure robust across a nine-point sensitivity sweep inside the pre-committed DESI criterion. See the V8.5 and V8.6 companion notes for the full derivations.

14.2 The interpolating function. *Substantially addressed in V8.4 revised.* The functional family $\nu(x) = \sqrt{(1 + \lambda/x)}$ is now derived as a theory-motivated result from the screen-energy construction rather than borrowed from MOND, with morphology-dependent λ (disks clustering near 3π/8, bulge-rich systems near 4/3) confirmed against SPARC at $p = 0.0016$ via partial correlation tests controlling for additional morphological covariates. See V8.4 revised for the full derivation. The remaining open direction is the theoretical derivation of the morphology-to-λ mapping from first principles.

14.3 The 13% gap in a_0. The predicted a_SFT = 1.05×10^{-10} m/s^2 is 13% below MOND's empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. Candidate explanations include QCD binding energy contributing to the effective Compton frequency (the proton's mass is approximately 99% binding energy, not rest mass of constituent quarks), cosmological evolution of H_0 between the epoch when a_0 was calibrated and z = 0, systematic uncertainties in the empirical determination of a_0 from galaxy samples, or the choice of interpolating function.

14.4 The second 2π from first principles. Section 6 identifies the second factor of 2π in the $4\pi^2$ decomposition as the ratio between Bekenstein-Hawking boundary normalization and Einstein bulk field equation coupling. This identification is correct algebraically but does not yet constitute a first-principles derivation. Showing that apparent-horizon thermodynamics requires this specific normalization ratio — rather than simply observing that the two conventions differ by 2π — would complete the geometric story.

14.5 Cosmological sector parameters. *Substantially addressed in V8.5.* The cosmic web state weights are now derived from axis factorization rather than fit from data, and the static wall and filament projection corrections are derived from spherical-screen second-moment integrals. The

earlier bimodal fit parameters ($\beta = 1.5$, $\mu = 3.5–4.0$) are superseded by the forced ladder framework. Remaining direction: further narrowing of any residual parameter dependences within the V8.5 ladder structure.

14.6 Cross-simulation validation. All cosmological results currently use IllustrisTNG TNG300 exclusively. Validation against independent cosmological simulations (EAGLE, SIMBA, Magneticum) with different sub-grid physics would confirm that the derived ladder and projection corrections are physical results rather than TNG300 artifacts.

14.7 Beyond FLRW minisuperspace. *Partially addressed in V9.* The weak-field lag-action program produced a first-post-Newtonian candidate sector (Branch C*) that passes the classic PPN β gate and the Eddington light-deflection gate through exact coefficient matching on the exponential metric form $g_{tt} = -\exp(-2U)$, $g_{rr} = \exp(2U)$. The full nonlinear field theory extension remains open, as does the derivation of the Goal 3 source-sector bridge. See the V9 companion note for the current status and the phase-gradient clock-field bridge candidate, which has been quantitatively constrained by recombination-scale mildness bounds but not yet derived from canonical V8 structure.

14.8 Passive channel microphysics. *Partially addressed through V8.5's forced ladder derivation*, which establishes axis factorization as the mechanism that forces specific state weights. The deeper question of why virialization specifically is the relevant criterion for screening contribution, and how that criterion is forced by the screen-energy formulation at the local level, remains an open direction for future work.

15. Conclusion

From the single postulate that every massive particle pays a continuous information-maintenance cost at its Compton frequency, with each complete phase cycle demanding one Planck area of cosmological boundary, we derive:

(1) A modified Friedmann equation $H^2 = 2G\rho/(3\pi)$, differing from the standard equation by exactly $1/(4\pi^2)$, derived algebraically (Section 3) and from the holographic count equation (Section 7) with zero free parameters.

(2) A galactic acceleration threshold $a_{SFT} = cH_0/(2\pi) = 1.05 \times 10^{-10}$ m/s^2, agreeing with MOND's empirical a_0 to 13% with zero free parameters, confirmed against 137 SPARC galaxies at 63% error reduction (Section 4).

(3) A unification of four cosmological appearances of $4\pi^2$—the Friedmann ratio, the MOND threshold, the holographic capacity gap, and the dark energy scale—as the product of two independent geometric identities: the phase convention (h vs $\hbar$) and the boundary-versus-bulk normalization (Bekenstein-Hawking vs Einstein coupling) (Sections 5–6).

(4) A holographic count equation and constrained minisuperspace action principle that serve as the native variational formulation of the theory (Sections 7–8).

(5) A dark energy equation of state ($w_0 = -0.837$, $w_a = -0.615$) matching DESI observations, with CMB compatibility at 0.078% RMS, derived from a bimodal entropy ledger using IllustrisTNG simulation data (Section 11).

(6) Exact recovery of the Schwarzschild exterior metric and TOV hydrostatic equations in the strong-field limit (Section 12).

(7) Falsifiable predictions for environment-dependent structure growth (5–6% split at $z < 0.5$) and anomalous ISW signals in supervoids, testable by DESI, Euclid, and CMB-S4 (Section 13).

(8) A weak-field lag-action candidate sector (Branch C*) that passes the classical post-Newtonian gates — $\beta = 1$ exactly from exponential-metric coefficient matching, and first-order light deflection of $4GM/(bc^2)$ matching the Eddington result — with a quantitatively constrained phase-gradient source-sector bridge documented in the V9 companion note.

The dark energy problem, the MOND acceleration scale, and the holographic capacity gap are not three separate mysteries. They are three measurements of the same quantity: the $(2\pi)^2$ geometric factor between angular and linear units of quantum information, projected onto a spherical cosmological boundary through the normalization conventions of holographic thermodynamics and bulk gravitational physics. This identification requires no new fields, no new particles, and no modification to general relativity. It requires only that existence is computational, and that the universe measures computation in complete phase cycles.

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